

Did COVID-19 Change Safe-Haven Behaviour?

Evidence from Volatility, Dynamic Correlations, Structural Breaks, and Regime Switching

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Abstract

This paper examines whether COVID-19 changed the risk dynamics and safe-haven properties of traditional defensive assets. Using daily data from 1990 to 2026, we study the relationship between the S&P 500, Eurostoxx 50, and four safe-haven candidates: gold, US 10-year Treasuries, the Swiss franc, and the Japanese yen. The analysis combines GJR-GARCH models, Dynamic Conditional Correlations, Bai-Perron structural break tests, Markov-switching models, and Equal Risk Contribution portfolios. The results suggest that COVID-19 did not eliminate safe-haven behaviour, but made it more unstable, asset-specific, and regime-dependent. We find structural breaks in all safe-haven/equity pairs and evidence of changes in high-correlation breakdown regimes. Portfolio results also show that relying on pre-COVID allocations slightly increases post-COVID volatility. Overall, traditional safe havens remain useful, but investors should no longer treat them as stable or interchangeable across regimes.

Keywords: COVID-19, safe-haven assets, GJR-GARCH, DCC correlations, structural breaks, Markov-switching, ERC portfolio

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1 Introduction

The COVID-19 pandemic was one of the largest financial shocks since the 2008 Global Financial Crisis. Besides its impact on the real economy, the pandemic also strongly affected financial markets. During this period, markets experienced very high volatility, liquidity problems and sudden changes in investor sentiment. In March 2020, equity markets recorded large drawdowns, while traditional diversification strategies became less effective because correlations between asset classes increased significantly. At the same time, governments and central banks introduced exceptional fiscal and monetary measures, which changed the macro-financial environment in which financial markets operate.

These changes pose an important question about the impact of the pandemic on risk management and portfolio construction: has there been a change in the nature of financial risk since COVID-19? Specifically, did the relationship between risky and safe assets remain the same post-COVID? If so, diversification benefits may have become weaker.

Several stylized facts of financial markets are well documented in the literature. Financial returns often exhibit volatility clustering, fat tails and time-varying correlations (Bollerslev, 1986; R. Engle, 2002; R. F. Engle, 1982). Previous studies also show that correlations between risky assets generally increase during crisis periods, reducing diversification benefits exactly when investors need them most (Longin & Solnik, 2001). In addition, the literature on safe haven assets suggests that traditional defensive assets such as gold, sovereign bonds, the Swiss franc or the Japanese yen do not always provide stable protection during periods of market stress (Baur & Lucey, 2010). More recent studies related to COVID-19 also indicate that the pandemic may have changed volatility regimes, contagion dynamics and cross-asset relationships (Corbet et al., 2020).

This paper studies whether the post-COVID period is associated with structural changes in financial risk dynamics. In particular, we investigate whether volatility persistence, cross-asset correlations and the relationships between equity markets and traditional safe-haven assets changed after the pandemic. Unlike studies imposing arbitrary break dates, we use Bai and Perron (2003) structural break tests to identify regime changes directly from the data. Our results identify a major structural break around the end of February 2020 for US equity markets.

Using a dataset composed of major equity indices and traditional safe haven assets, we analyze volatility persistence, dynamic correlations, structural breaks and regime behavior before and after the pandemic. To do so, we use several econometric methods commonly applied in empirical finance and risk management, including GJR-GARCH models with Student-t innovations, Dynamic Conditional Correlation (DCC) models, structural break tests and Markov-switching models. We also examine the portfolio implications of these changes through Equal Risk Contribution (ERC) allocations and dynamic risk exposures.

Our results show that there have been some modifications in the financial risk structure in the post-pandemic era compared to the pre-pandemic era. The evidence suggests the occurrence of higher volatility values, greater instability of cross-asset correlations, as well as the emergence of more switches into adverse correlations after the pandemic. The analysis reveals that some of the previously existing safe haven asset relationships weakened post-pandemic, resulting in the weakening of the portfolio diversification effect. Moreover, the use of the pre-COVID portfolio weights after the pandemic resulted in somewhat higher portfolio volatility, implying that the efficiency of such a portfolio strategy declined after the pandemic occurred.

The rest of the paper is organized as follows. Section 2 reviews the literature and presents the main hypotheses of the study. Section 3 describes the data. Section 4 explains the econometric

methodology. Section 5 presents the empirical results, while Section 6 discusses their economic and portfolio implications. Section 7 concludes.

2 Literature Review

The literature on financial risk dynamics mainly focuses on volatility persistence, changing correlations, contagion effects, safe haven behavior and structural changes in market regimes. Overall, these studies suggest that crisis periods can significantly modify the way risk is transmitted across financial markets and reduce the effectiveness of diversification strategies.

A first part of the literature studies volatility dynamics and the main stylized facts of financial returns. Financial returns are known to exhibit excess kurtosis, volatility clustering and time-varying volatility. R. F. Engle (1982) introduced the ARCH model to capture conditional heteroskedasticity, while Bollerslev (1986) extended this framework with the GARCH model. These models show that volatility is persistent and tends to cluster during crisis periods. Later studies also introduced asymmetric volatility models such as the GJR-GARCH specification, which shows that negative shocks usually have a stronger effect on volatility than positive shocks. In addition, Student-t distributions are often used because financial returns are generally not normally distributed and display fat tails. Bollerslev (1986) also highlights that GARCH-type models are particularly useful to capture the leptokurtic behavior of financial returns, which became even more visible during the COVID-19 crisis. More recent studies, such as Vera-Valdés (2022), suggest that COVID-19 modified volatility persistence and reduced the effectiveness of standard forecasting models when structural breaks are ignored.

Another part of the literature looks at how correlations between financial assets evolve during crisis periods. Longin and Solnik (2001) show that correlations between international stock markets tend to rise during market downturns, making diversification strategies less effective in bad market conditions. They also find that this effect is stronger during bear markets than during normal periods. Forbes and Rigobon (2002) study contagion during financial crises and explain that higher correlations do not always mean true contagion, since they can also come from higher market volatility and stronger interdependence between markets. Later, R. Engle (2002) developed the Dynamic Conditional Correlation (DCC) model, which makes it possible to study correlations that vary over time instead of assuming constant correlations. Since then, DCC-GARCH models have often been used to analyze the evolution of cross-asset relationships and diversification benefits over time.

Another strand of the literature studies the role of safe haven assets during crisis periods. Baur and Lucey (2010) show that gold may act as a safe haven during some crisis periods, although this property is not always stable over time and may only last for a short period after the initial shock. Baur and McDermott (2010) extend the analysis to different countries and find that safe haven characteristics can vary depending on the market and the type of crisis considered. More generally, several studies suggest that traditional defensive assets such as sovereign bonds, the Swiss franc or the Japanese yen may become less effective during periods of strong market stress or structural instability. In the specific context of COVID-19, Corbet et al. (2020) also suggest that some assets traditionally considered as safe havens may have transmitted financial stress instead of protecting portfolios.

Finally, another strand of the literature focuses on structural breaks and regime changes in financial markets. Bai and Perron (2003) propose econometric methods to identify multiple structural breaks in time series data without imposing predefined break dates. Hamilton (1989) introduces Markov-switching models, which allow financial markets to switch between different

regimes with different statistical properties. These models are particularly useful in the context of COVID-19 because the pandemic may have changed volatility dynamics, correlations and investor behavior in a persistent way. Recent COVID-related studies also suggest that the pandemic increased contagion effects, financial uncertainty and instability in diversification patterns across global markets (Corbet et al., 2020; Zhang et al., 2020).

Based on this literature, this paper investigates whether the post-COVID period is associated with structural changes in financial risk dynamics. More specifically, we test three main hypotheses:

- **H1:** Volatility persistence and tail risk increased after the COVID-19 shock.
- **H2:** Safe haven relationships between gold, US Treasuries, USDCHF, USDJPY and equity indices experienced structural breaks after COVID-19, reducing their hedging effectiveness.
- **H3:** Higher-correlation regimes identified through Markov-switching models reduced the stability and efficiency of ERC portfolio allocations after 2020.

By combining volatility models, dynamic correlation analysis, structural break tests and regime-switching approaches, this study aims to evaluate whether COVID-19 significantly changed the structure of financial risk.

3 Data

The initial dataset provided contains daily observations on 14 assets spanning from December 31, 1989 to April 24, 2026. For the purpose of this study, we keep only six assets : two equity benchmarks, the S&P 500 and the Eurostoxx 50, and four assets considered as safe-haven, Gold (spot price in USD), the US 10-year Treasury yield, and two exchange rates, USDCHF and USDJPY.

Log-returns are computed directly from raw prices, before any forward-filling, to avoid introducing artificial zero-return observations on non-trading days that would distort our GARCH estimates and autocorrelation diagnostics. The transformation applied differs by asset. For the S&P500, Eurostoxx 50, and Gold, we use the standard log return $r_t = \ln(P_t/P_{t-1})$. For the US 10-year yield, we apply a duration approximation, $r_t = -\Delta y_t \times D_{\text{mod}}/100$ so that a positive value of r_t corresponds to a bond price increase. We use a modified duration of 8.5 years as an approximation for a 10-year US Treasury bond. For USDCHF and USDJPY, we negate the log-return, $r_t = -\ln(P_t/P_{t-1})$, so that an appreciation of the Swiss Franc or Japanese yen represents a positive return. These sign conventions ensure that a positive value represents a profit for the holder across all six series, which facilitates interpretation. Returns are multiplied by 100 to be expressed in percentage point, improving numerical stability in our estimation models. After computing returns and dropping the first observation, any remaining rows with missing values are removed.

The cleaned dataset covers the period from January 3, 1990 to April 24, 2026, with a total of 8476 observations. The pre/post COVID split is set at March 1, 2020, corresponding to the World Health Organization's official declaration of a global pandemic on March 11, 2020 with a 10-day anticipation margin that better reflects the timing of the initial market shock. This date is later supported by the structural break analysis in Section 5. The pre-COVID sub-sample contains 7038 observations, and the post-COVID subsample contains 1438 observations. For analysis comparing pre- and post-COVID parameters (GJR-GARCH, Markov Switching), the

pre-COVID window is further restricted to start on January 1, 2010, to exclude the 2008 Global financial Crisis from the pre-COVID baseline. This reduces the risk that results are driven by the 2008 Global Financial Crisis rather than by the COVID-19 shock.

There are a number of limitations associated with the data which need to be mentioned. For instance, the US-10 year bond returns are just proxies using the approximation of the bond duration which is assumed to be constant throughout the entire sample at 8.5 years although duration is variable depending on the yield change. Also, as convexity is not included in the approximation, its impact is disregarded, especially when there is a large movement in yields. The result will be a lower estimation of bond price increases and a higher one for bond price decreases, especially in situations like in 2022. Another limitation of the data used is the absence of any consideration of transaction costs or liquidity issues.

4 Methodology

This section describes the empirical methods used to test the three hypotheses stated in Section 2. The analysis consists of five interdependent parts : preliminary diagnostics, univariate volatility modeling, multivariate dependence, structural breaks and regime detection, and portfolio implications.

4.1 Sample Periods

The analysis uses two sample periods. First of all, the entire period from 1990 through 2026 is taken into account when computing descriptive statistics, modeling volatility, and dynamic correlations, since a longer data series ensures a better estimation of these indicators. When making direct comparisons between pre- and post-COVID parameters, such as the GARCH estimates or regimes, a shorter sample will be considered, starting from 2010. By doing this, the extreme volatility that occurred during the Global Financial Crisis of 2008 can be excluded from the analysis.

4.2 Preliminary Diagnostics

Before fitting any model, we run a set of diagnostic tests on the return series to verify that the modeling assumptions are satisfied and to motivate the choice of GJR-GARCH with Student-t innovations.

Normality testing uses the Jarque-Bera and Kolmogorov-Smirnov tests. Failure of normality for all six series indicates fat tails, thus making it necessary to employ Student-t distribution instead of Gaussian innovations. Stationarity is verified using the ADF and KPSS tests jointly. All six series were found to be stationary, an important condition for validity of the statistical results obtained from the model. The final test is ACF of Squared Returns and Ljung Box Test for volatility clustering.

We also compute descriptive statistics for each asset across the pre- and post-COVID subsamples.

4.3 Univariate Volatility Modelling: GJR-GARCH

We model the time-varying volatility of each asset using a GJR-GARCH(1,1) with Student-t innovations. Returns decompose as:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \quad (1)$$

The conditional variance follows:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}_{\{\varepsilon_{t-1} < 0\}} + \beta \sigma_{t-1}^2 \quad (2)$$

The asymmetry parameter γ captures the leverage effect: negative shocks raise future volatility more than positive ones of equal size. Persistence is measured by $\alpha + \gamma/2 + \beta$. The Student-t distribution accommodates the fat tails confirmed in Section 4.2.

Estimation of the model is based on three different samples: full, pre-COVID, and post-COVID. Full-sample estimate of conditional volatility $\hat{\sigma}_t$ serves as an input in the DCC model. Comparison of estimates before and after COVID-19 is done to test if there was any structural change in the volatility behavior of assets during the period, using a Wald test on the equality of persistence between the two samples. The adequacy of the estimated DCC models is verified using the Ljung-Box test on the squared standardized residuals $z_t^2 = \varepsilon_t^2 / \hat{\sigma}_t$. Value of p-statistic greater than 0.05 indicates that the ARCH effects have been incorporated. Five out of six series pass this test while the sixth series, gold, fails the test.

In the final step, we perform the sup-F test by Andrews (1993) for $\hat{\sigma}_t$ over the post-2010 period. Rejection of the no break null for all assets proves that volatility shows structural instability over the period, which calls for further analysis of correlation instability in Section 4.5.

4.4 Dynamic Conditional Variance

To study the time-varying co-movement between safe haven assets and equity indices, we use the DCC(1,1) model of Engle (2002). The estimation proceeds in two steps.

In the first step, the standardized residuals from the GJR-GARCH model are extracted:

$$u_{i,t} = \frac{\hat{\varepsilon}_{i,t}}{\hat{\sigma}_{i,t}} \quad (3)$$

These residuals have each asset's individual volatility dynamics removed, leaving only the cross-sectional dependence structure.

In the second step, the time-varying correlation matrix is modelled as:

$$Q_t = (1 - a - b) \bar{Q} + a u_{t-1} u_{t-1}^\top + b Q_{t-1} \quad (4)$$

where $\bar{Q} = \text{Cov}(u_t)$ is the unconditional covariance matrix of the standardized residuals, and $a, b > 0$ with $a+b < 1$ are estimated by maximum likelihood. The conditional correlation matrix is then obtained by rescaling:

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2} \quad (5)$$

We test each safe haven asset against both equity benchmarks (S&P 500 and Eurostoxx 50), yielding eight correlation pairs (4 safe havens x 2 benchmarks). This dual-benchmark design allows us to assess whether changes in hedging properties differ across US and European equity markets.

To complement the DCC correlations, we compute a conditional beta decomposition for each safe haven/equity pair, defined as:

$$\beta_t = \rho_t \times \frac{\hat{\sigma}_{sh,t}}{\hat{\sigma}_{m,t}} \quad (6)$$

which separates the correlation component from the relative volatility component. This decomposition allows us to assess whether changes in hedging effectiveness are driven by shifts in correlation, in relative volatility, or in both.

Finally, we also compute rolling 252-day correlations between each safe haven and equity pair. While less statistically rigorous than the DCC, this approach provides an intuitive visual confirmation of the time-varying dependence structure and does not rely on any distributional assumptions.

4.5 Structural Breaks Regimes

To test whether the DCC correlations underwent regime changes around COVID-19, we apply a two-stage structural break procedure to each of the eight safe haven/equity correlation pairs.

In the first stage, the sup-F test of Andrews (1993) tests the null hypothesis of no structural break against the alternative of exactly one break at an unknown date, with rejection indicating that the correlation series is not stable over time. If the sup-F test fails to reject, we report no statistical evidence of a break for that pair.

At the next step, for all those pairs for which the sup-F statistic indicates that the null hypothesis should be rejected, the Bai and Perron (2003) test for multiple structural breaks. We restrict the number of potential breaks to no more than three, based on three main macroeconomic changes that occurred over the observed time span: the dot-com boom, the Global Financial Crisis of 2008, and COVID-19. In order to find the globally best set of break dates for each considered number of breaks $m = 0, 1, 2, 3$, one needs to minimize the total sum of squared residuals. The final number of breaks is determined using the Bayesian Information Criterion (BIC).

4.6 Markov-Switching model

To characterize the regime dynamics of each correlation pair, we estimate a two-state Markov-switching model (Hamilton, 1989) on each of the eight DCC correlation series. The two states capture a normal hedging regime, characterized by low correlation, and a breakdown regime, characterized by high correlation where the safe haven provides little protection.

The regime-specific mean, variance, and a 2x2 transition probability matrix are estimated by the model where the diagonal elements p_{11} and p_{22} indicate the probability of remaining in the same regime. The expected length of stay in the regime i can be computed using $1/(1 - p_{ii})$.

The model is estimated separately on the pre- and post-COVID subsamples. A Wald test on the equality of the transition parameters across the two periods is used to assess whether COVID-19 changed the regime dynamics. An increase in the expected duration of the breakdown regime, or a rise in its unconditional probability after 2020, would indicate a persistent deterioration in safe haven effectiveness.

4.7 Equal Risk Contribution (ERC) Portfolio

In order to express the implications of our statistical analysis in terms of a portfolio, we calculate equal risk contribution (ERC) weights separately for the pre- and post-COVID periods. In the ERC portfolio, the weights are calculated so that each investment is assigned an equal share in total portfolio risk, based solely on the covariance matrix.

To quantify the economic consequences of the post-COVID covariance shift, we measure the cost of inaction, which is the additional annualized volatility borne by an investor who keeps pre-COVID weights in the post-COVID environment:

$$\text{Cost} = \frac{q}{w_{\text{pre}}^T \Sigma_{\text{post}} w_{\text{pre}}} - \frac{q}{w_{\text{post}}^T \Sigma_{\text{post}} w_{\text{post}}} \quad (7)$$

A positive cost indicates that the pre-COVID allocation is no longer optimal once the new risk structure is taken into account.

5 Empirical Results

This section examines whether safe-haven behavior in financial markets changed after COVID-19. Overall, the results suggest a partial but economically meaningful change, rather than a simple shift from a stable pre-2020 environment to a weaker post-COVID one. Looking across the descriptive statistics, GJR-GARCH estimates, Wald tests, volatility break tests, DCC and rolling correlations, Bai-Perron breaks, Markov-switching estimates, and ERC allocation results, the same pattern appears: defensive properties have not disappeared, but they have changed across assets and benchmarks.

5.1 Post-COVID Market Conditions and Univariate Risk Dynamics

The descriptive statistics results already indicate that the market environment confronting risky assets and candidate havens changed materially after March 2020, and Figure 1 shows the same point dynamically through the full-sample GJR-GARCH conditional volatilities. The March 2020 shock appears immediately in the equity series, but the post-COVID period is not just a short-lived spike. Gold, Treasuries, and the currency havens all show new volatility episodes after 2020, especially around the 2022 repricing of interest rates and exchange rates. This is important for the safe-haven question because a hedge is useful only if its own risk remains reasonably stable.

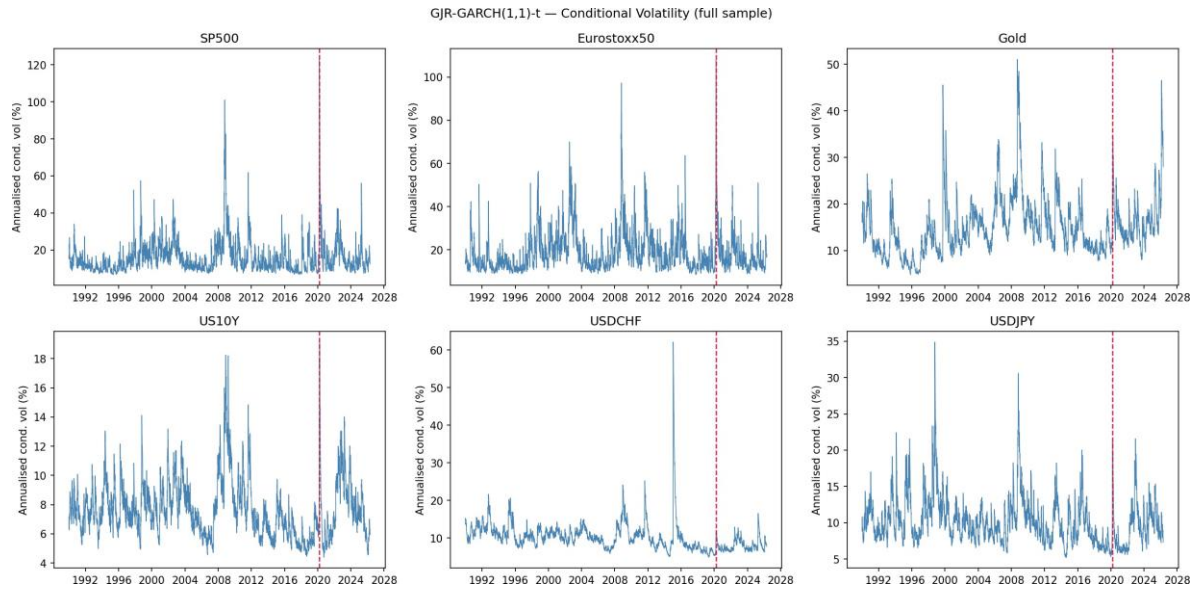


Figure 1: Full-sample GJR-GARCH conditional volatilities. The red dashed line marks the WHO pandemic declaration on March 11, 2020.

The descriptive-statistics results are also heterogeneous in a way that matters economically. The annualized mean return on gold increased from 5.3% before COVID to 17.3% after March 2020. The Swiss franc also had a higher annualized mean return after the start of the pandemic (4.5% compared to 2.5%). By contrast, US 10-year Treasuries and the Japanese yen record negative average post-COVID returns, at -2.6% and -4.5% respectively. These numbers should not be read as a safe-haven ranking, since a safe haven is defined by its co-movement with equities during stress, not by its unconditional return. They do, however, show that the post-COVID environment was not uniformly supportive of the assets usually treated as defensive.

Turning to the GJR-GARCH estimates, the picture on volatility persistence is more nuanced. Volatility persistence remains very high in both subsamples for all six assets, and the persistence Wald-test results do not reject equality across the COVID break for any asset, even though the post-COVID point estimates are lower for gold, Treasuries, and especially the Swiss franc. At the same time, the volatility sup-F results reject stability for every asset, with break dates ranging from February 25, 2020 for the S&P 500 to March–April 2022 for US Treasuries and the yen, and November 2, 2023 for gold.

There is conflicting evidence regarding hypothesis H1. The sup-F rejections and the emergence of new volatility episodes after 2022 clearly show that the post-COVID period was not simply a return to normal. However, when it comes to the Wald test results, nothing much changed, since none of the six assets exhibit a regime switch in the level of volatility persistence. The change following COVID-19 should be interpreted more as frequent and extreme volatility surges than as a change in the volatility regime itself. H1 is valid for tail risks and levels, but not for persistence. The more important change is likely in the cross-asset dependence structure, which is what the next section examines.

5.2 Structural Breaks in Safe-Haven Dependence

We now turn from individual volatility dynamics to the cross-asset dependence structure, which is what directly determines safe-haven effectiveness. The dynamic-correlation results are the

main evidence for the safe-haven analysis. The DCC estimates are very persistent, with $a + b = 0.9944$, so the changes in correlation should not be treated as short-lived noise. Figure 2 compares the DCC correlations with rolling 252-day correlations for the eight safe-haven/equity pairs. The results show that post-COVID breakdown episodes differ a lot across assets. The clearest case is US Treasuries against the S&P 500: after the first pandemic hedge episode, the correlation moves back toward zero and becomes positive several times from 2021 onward. Gold also looks less consistently defensive after 2020, as its correlation with equities turns positive at several points and often stays close to zero after the first pandemic shock. The Swiss franc also has periods of positive correlation with both equity benchmarks between 2021 and 2023. The yen moves back from very negative pandemic correlations toward values closer to zero, with some positive episodes as well. The rolling correlations show the same broad pattern, although the timing is naturally smoother.

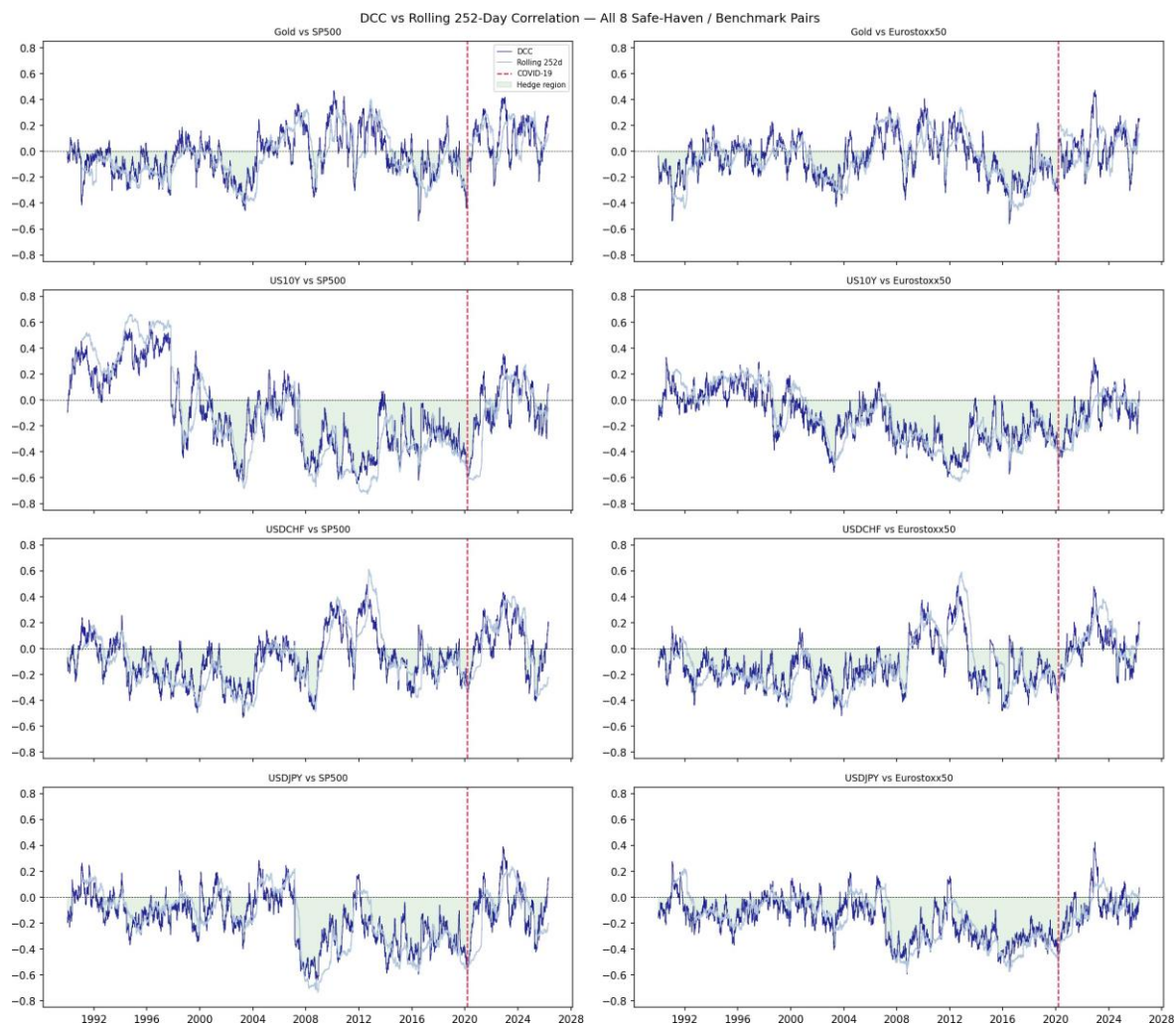


Figure 2: DCC and rolling 252-day correlations for the eight safe-haven/equity pairs. Negative correlation corresponds to stronger hedge or safe-haven behavior.

Figure 3 gives the corresponding conditional beta decomposition against the S&P 500. This distinction is important because a safe-haven beta can move for two reasons: the correlation with equities can rise, or the safe asset's own volatility can rise relative to equity volatility. The post-COVID evidence therefore cannot be interpreted from correlations alone. For Treasuries and the currency havens in particular, changes in relative volatility amplify or dampen the

correlation signal, so portfolio hedge ratios change even when the sign of the correlation is temporarily stable. The correlation signal alone, however, does not fully capture changes in hedging effectiveness.

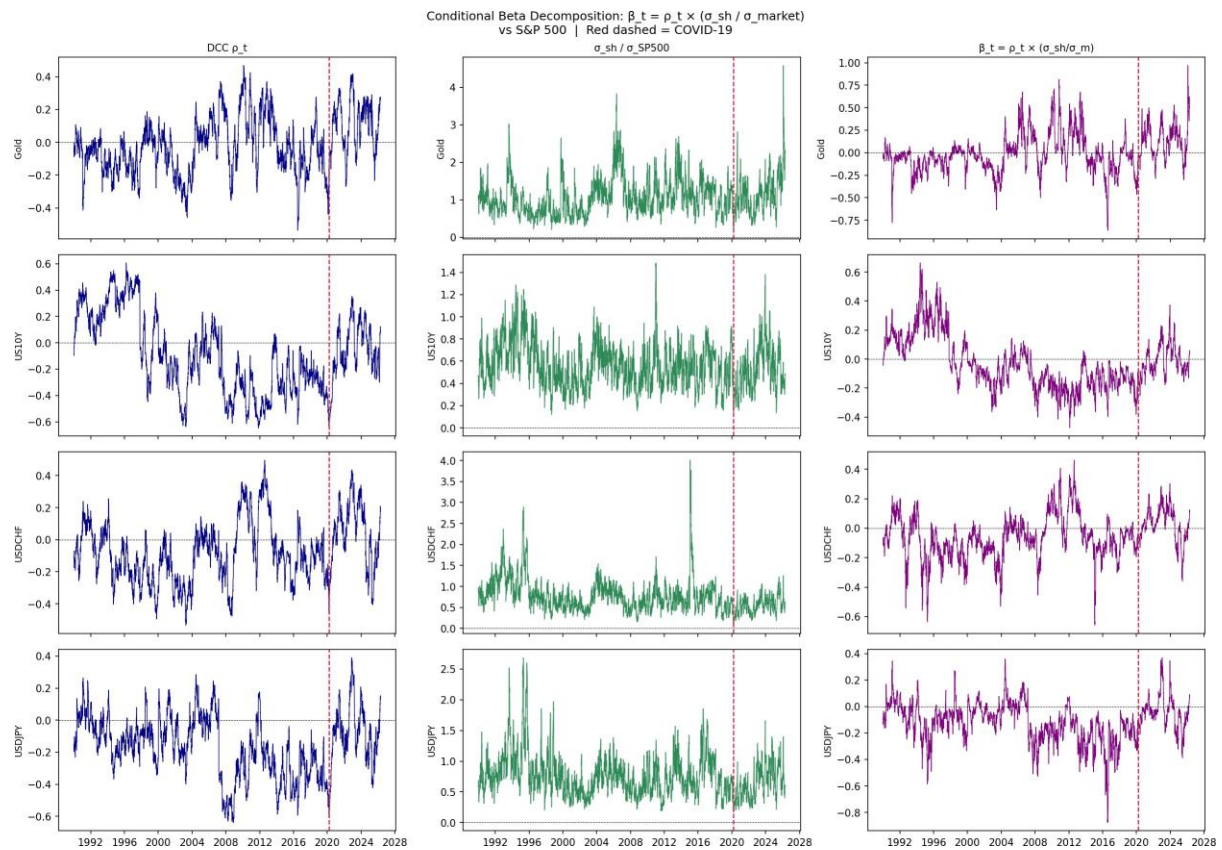


Figure 3: Conditional beta decomposition against the S&P 500: DCC correlation, relative volatility, and implied safe-haven beta.

Beyond the visual evidence, formal break tests confirm this instability statistically. We reject the hypothesis that there has been no break for each pair with the sup-F test by a large margin, and we identify three breaks for each pair using the Bai-Perron tests. Table 1 reports the final break for each safe-haven asset against both equity benchmarks, since that last break is the one most directly relevant to the COVID question. The dates fall within the interval from March 12, 2020 through April 8, 2021.

Table 1: Final COVID-era Bai-Perron break date by safe-haven/equity pair

Safe-haven asset	vs S&P 500	vs Eurostoxx 50
Gold	2020-08-04	2020-03-12
US 10Y Treasuries	2021-02-26	2021-04-08
Swiss franc	2020-08-18	2020-09-24
Japanese yen	2020-08-18	2021-01-26

All eight pairs first reject the no-break null in the sup-F test, and Bai-Perron selects three breaks in every case. The table reports the final break, i.e. the one associated with the COVID and immediate post-COVID period.

The timing of the breaks is also useful for interpretation. Gold shows the first COVID-era break

against Eurostoxx 50 on March 12, 2020, while the break against the S&P 500 comes later, on August 4, 2020. The Treasury pairs shift even later, in February and April 2021. This suggests that the change in bond hedging was not only linked to the first lockdown shock. It also seems connected to the period that followed, when inflation pressures and monetary tightening started to change the usual bond-equity relationship.

Results of the break tests indicate that H2 is valid: all pairs reject the no-break hypothesis with break dates coinciding with the date of the COVID crisis. Importantly, results from beta decomposition indicate that there was an adjustment in the hedge ratio even if correlation signs remained intact.

5.3 Regime Dynamics After COVID

Break dates alone do not tell us whether the underlying hedging states themselves changed. The Markov-switching estimates show that they did. For every pair, the Wald test comparing the full pre-COVID and post-COVID regime parameter vectors rejects stability with a p-value below 0.0001. This is a stronger result than a change in average correlation: it implies that the persistence and composition of the low-correlation and high-correlation states were themselves altered by the post-COVID environment. The smoothed regime probabilities in Appendix Figure A1 provide the visual counterpart to this result.

Table 2 reports how long the high-correlation “breakdown” regime is expected to last before and after COVID, based on the Markov-switching estimates and the Wald tests. The clearest deterioration is for US Treasuries against the S&P 500, where the expected duration of this bad regime rises from 91.5 to 135.6 trading days. For a US equity investor, this is not what we would normally expect from a safe-haven asset. Gold gives a more mixed picture: it remains almost unchanged against the S&P 500, moving from 109.6 to 104.7 days, but improves more clearly against Eurostoxx 50, where the duration falls from 177.8 to 97.8 days. The yen also improves against both benchmarks, while the Swiss franc becomes sharply more effective against Eurostoxx 50 but remains fragile against the S&P 500, with the bad regime lasting slightly longer after COVID than before.

Table 2: Expected duration of the high-correlation breakdown regime (days)

Pair	Pre-COVID	Post-COVID	Wald p -value
Gold vs S&P 500	109.6	104.7	< 0.0001
Gold vs Eurostoxx 50	177.8	97.8	< 0.0001
US 10Y vs S&P 500	91.5	135.6	< 0.0001
US 10Y vs Eurostoxx 50	155.2	80.4	< 0.0001
CHF vs S&P 500	210.3	215.7	< 0.0001
CHF vs Eurostoxx 50	183.0	45.6	< 0.0001
JPY vs S&P 500	105.6	71.6	< 0.0001
JPY vs Eurostoxx 50	71.1	53.3	< 0.0001

Durations are computed as $1/(1 - p_{ii})$ from the two-state Markov-switching model, where the breakdown regime is the high-correlation state in which hedging is weakest.

This heterogeneity is important for interpreting the results. Post-COVID safe haven behavior becomes more specific by benchmark and dependent on states. While gold and the yen remain relatively robust and the Swiss franc becomes region-specific, Treasuries lose reliability precisely in the US equity pairing where their historical reputation is strongest.

Overall, the Wald test rejection is uniform across all pairs, confirming that COVID altered not just average correlations but the persistence and composition of the underlying regimes. H3 is supported, but the practical implication is asset-specific rather than systemic.

5.4 Portfolio Implications of the New Safe-Haven Structure

The covariance shifts documented in 5.2 and 5.3 have direct portfolio implications. As shown in Table 3, the ERC allocation after COVID gives less weight to gold and especially to US Treasuries, while increasing the weights of the Swiss franc and the yen. Since the ERC portfolio is based only on the covariance matrix, this shift does not come from return forecasts but from the change in risk and co-movement across assets. In practical terms, a risk-balanced investor would rely less on the traditional bond-heavy safe-haven allocation once the post-COVID covariance structure is taken into account.

Table 3: Equal-risk-contribution weights before and after COVID

Asset	Pre-COVID	Post-COVID
S&P 500	0.1334	0.1101
Eurostoxx 50	0.1092	0.1114
Gold	0.1202	0.1042
US 10Y Treasuries	0.2988	0.2606
Swiss franc	0.1542	0.2172
Japanese yen	0.1842	0.1964

Using pre-COVID ERC weights in the post-COVID covariance matrix raises annualized portfolio volatility from 6.96% to 7.16%, implying a cost of inaction of 0.19 percentage points.

The 0.19 percentage point cost of inaction is modest, but it shows that treating pre-COVID allocations as still valid is no longer costless. The ERC results confirm that the changes documented in 5.2 and 5.3 are not only statistically visible but economically meaningful. Defensive assets still work, but their relative usefulness has shifted in a way that is no longer costless for a target-risk investor.

6 Discussion

The post-COVID period is characterized by a more fragile and regime-dependent risk structure across all asset classes studied. The sup-F test rejects volatility stability for every series around March 2020, and the disruption did not stop at the initial shock : gold, Treasuries, and the currency havens all entered new volatility episodes in 2022 as central banks tightened aggressively. Pre-COVID risk models are therefore miscalibrated in two ways: they underestimate the level of volatility and miss the regime dependence that now characterises it, meaning Value-at-Risk estimates built on pre-2020 data will systematically understate tail risk.

In terms of diversification, the Bai-Perron tests identify a COVID-era structural break in all eight safe-haven/equity correlation pairs, while the estimates of the Markov-switching model confirm that the duration of the breakdown regime increased for most of these asset combinations. The worst situation concerns US Treasuries against the S&P 500, as the length of time of high-correlation rose from 91.5 to 135.6 trading days, meaning investors are exposed to a non-hedging environment for roughly nine additional weeks per episode compared to the pre-COVID baseline.

The degree of deterioration varies enough across assets that treating gold, Treasuries, the franc, and the yen as interchangeable defensive assets is no longer justified.

Regarding gold, one should be cautious about the interpretation of its DCC correlations. Indeed, its GJR-GARCH residuals, which are then standardized and used to feed the DCC, still carry a non negligible portion of ARCH structure, which is evidenced by the Ljung-Box test rejecting the null hypothesis of no remaining ARCH structure. These standardized residuals still carry volatility clustering, making the DCC estimated correlations distorted. Also, the fixed-duration estimation for the US Treasuries may have exaggerated the loss of hedging property of this asset class post COVID as during sharp yield increases, this assumption overstates the bond price loss, which in turn makes the correlation with equity higher than it truly is, thus making the hedging property look worse than it actually is.

These findings have practical consequences for portfolio management. Keeping pre-COVID ERC weights in the post-COVID covariance environment costs 0.19 annualized percentage points of excess volatility. The post-COVID ERC allocation shifts away from Treasuries and toward the Swiss franc and yen, not because return forecasts changed, but because the covariance structure did. The question for investors is no longer whether to hold safe havens, but which ones and in what regime.

7 Conclusion

This paper asked whether COVID-19 permanently altered the structure of financial risk and, more specifically, whether traditional safe havens still deliver the diversification benefits investors have historically relied on. The short answer is that they do, but less reliably than before. Structural breaks appear in all eight safe-haven/equity correlation pairs, the high-correlation breakdown regime lasts longer post-2020 for most assets, and the pre-pandemic assumption that gold, Treasuries, the Swiss franc and the Yen are broadly interchangeable defensive assets no longer holds.

These findings have concrete implications for risk management. Pre-COVID models are miscalibrated in two ways: they underestimate volatility levels and miss the regime-dependence that now characterises safe-haven correlations. The post-COVID optimal allocation shifts weight away from Treasuries and toward the Swiss franc and yen, not because return forecasts changed, but because the covariance structure did.

Two limitations qualify these findings. The Treasury return series is constructed via a fixed-duration approximation that overstates bond price losses when yields move sharply, which likely exaggerates the post-COVID deterioration in bond hedging effectiveness, particularly for the 2022 tightening episode. The post-COVID subsample also contains only 1,438 observations, reducing statistical power in the persistence tests and making some results indicative rather than definitive. As the post-pandemic period lengthens and more data accumulates, several of the borderline findings reported here will either solidify or dissipate, and the picture will sharpen accordingly.

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A Appendix

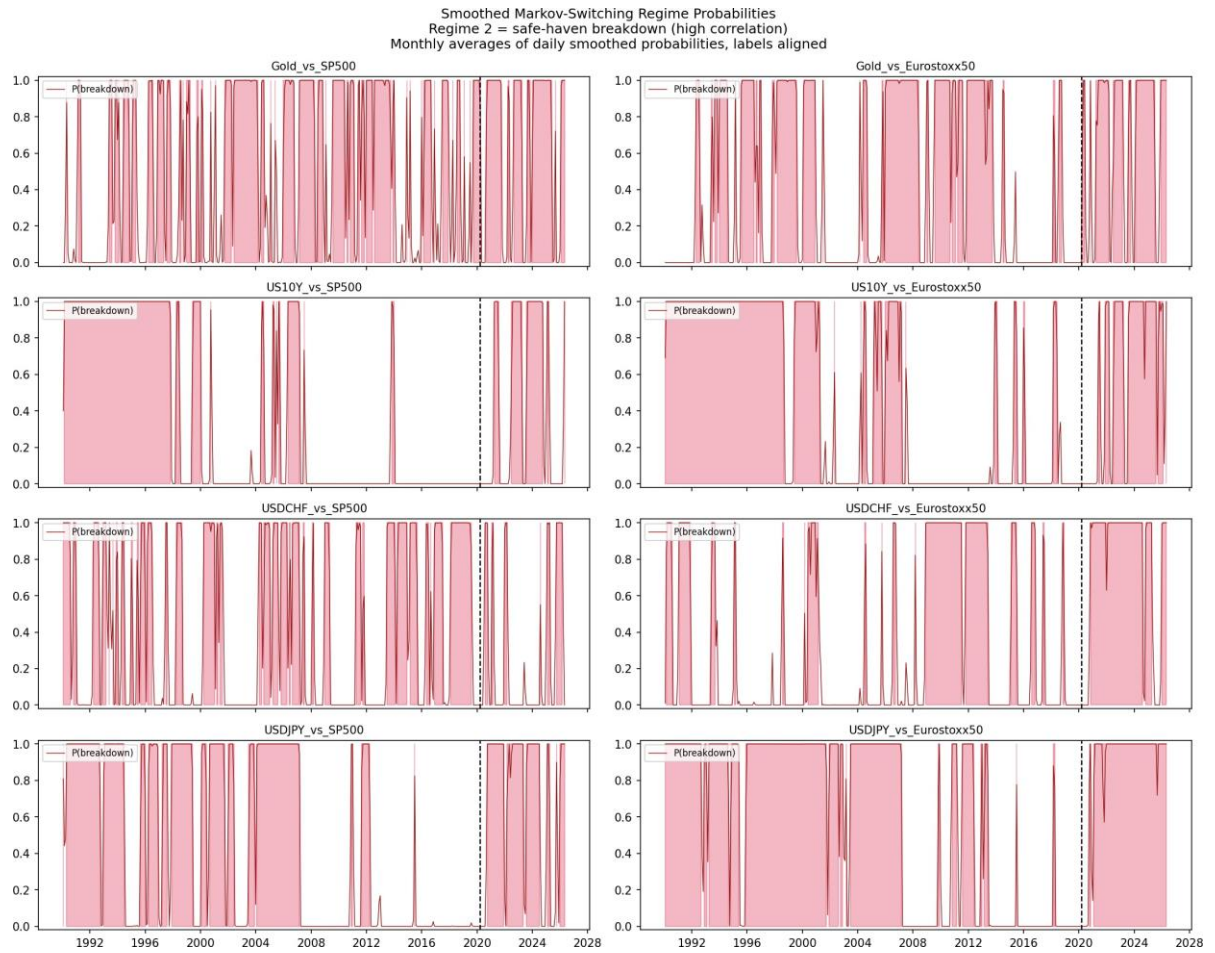


Figure A1: Smoothed Markov-switching probabilities of the high-correlation safe-haven breakdown regime.